Semantic Exploration Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Analysis of Lag-Based Semantic Distance

PENPAL Project

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Contents

1 Overview

The semantic exploration outputs now use the **current lag-based structure**:

- `k`: lag
- `distance`: mean semantic distance for that lag
- `agent`: `"interleaved"` for the full exchange sequence, or `author_1` / `author_2` for speaker-specific trajectories

The processed filename is still `semantic_exploration_binned.parquet`, but the current generator computes **lag-based distances**, not the older non-overlapping window-centroid bins.

This notebook therefore no longer assumes old turn-level exploration tables. Instead it analyses:

1. **Interleaved sequence exploration** across conditions
2. **Speaker-specific exploration** using recovered role metadata

Important comparison note:

- **Slot 1 / Slot 2** is the directly comparable cross-condition axis.
- **Human / AI** in Human-AI and **Starter-side / Non-starter-side** in same-type conditions are only used as within-condition descriptive views.

Primary cross-condition object. Per story, we compute the trapezoidal area under each slot's mean distance(k) curve over the intersection of k values present for both slots, and take the unsigned difference:

$$A_{\mathrm{semexp}}(i) \;=\; \left|\,\mathrm{AUC}_{\mathrm{slot}=1,\,i} - \mathrm{AUC}_{\mathrm{slot}=2,\,i}\,\right|.$$

This is label-invariant and comparable across conditions. The lag-binned LMM slot contrast and the log-distance slope view are retained as supplemental. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

2 Load Data

Table 1: Loaded semantic exploration rows by condition and agent view

condition	agent	n_stories	n_rows	min_k	max_k
Human-AI	author_1	100	754	1	8
Human-AI	author_2	100	754	1	8
Human-AI	interleaved	100	1000	1	10
Human-Human	author_1	31	248	1	8
Human-Human	author_2	31	248	1	8
Human-Human	interleaved	31	310	1	10
AI-AI	author_1	39	312	1	8
AI-AI	author_2	39	312	1	8
AI-AI	interleaved	39	390	1	10

3 Preprocessing

```
## Interleaved rows: 1700
```

```
## Speaker rows: 2628
```

4 Primary: Per-Story AUC Asymmetry

Table 2: Stories with per-story AUC computable in both slots

condition	n_stories
Human-AI	100
Human-Human	31
AI-AI	39

Table 3: Per-story |AUC(slot 1) - AUC(slot 2)| by condition

condition	n	mean_A	sd_A	median_A
Human-AI	100	0.228	0.174	0.195
Human-Human	31	0.213	0.175	0.186
AI-AI	39	0.215	0.161	0.181

```
## One-way ANOVA: A_semexp ~ condition
##              Df Sum Sq  Mean Sq F value Pr(>F)
```

```
## condition     2  0.007 0.003619   0.123  0.884
## Residuals   167  4.919 0.029454
##
## Kruskal-Wallis (robust): A_semexp ~ condition
##
##  Kruskal-Wallis rank sum test
##
## data:  A_semexp by condition
## Kruskal-Wallis chi-squared = 0.3529, df = 2, p-value = 0.8382
##
##
## Pairwise Wilcoxon (BH-adjusted):
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  df_auc_stories$A_semexp and df_auc_stories$condition
##
##             Human-AI Human-Human
## Human-Human 0.78     -
## AI-AI       0.78     0.78
##
## P value adjustment method: BH
```

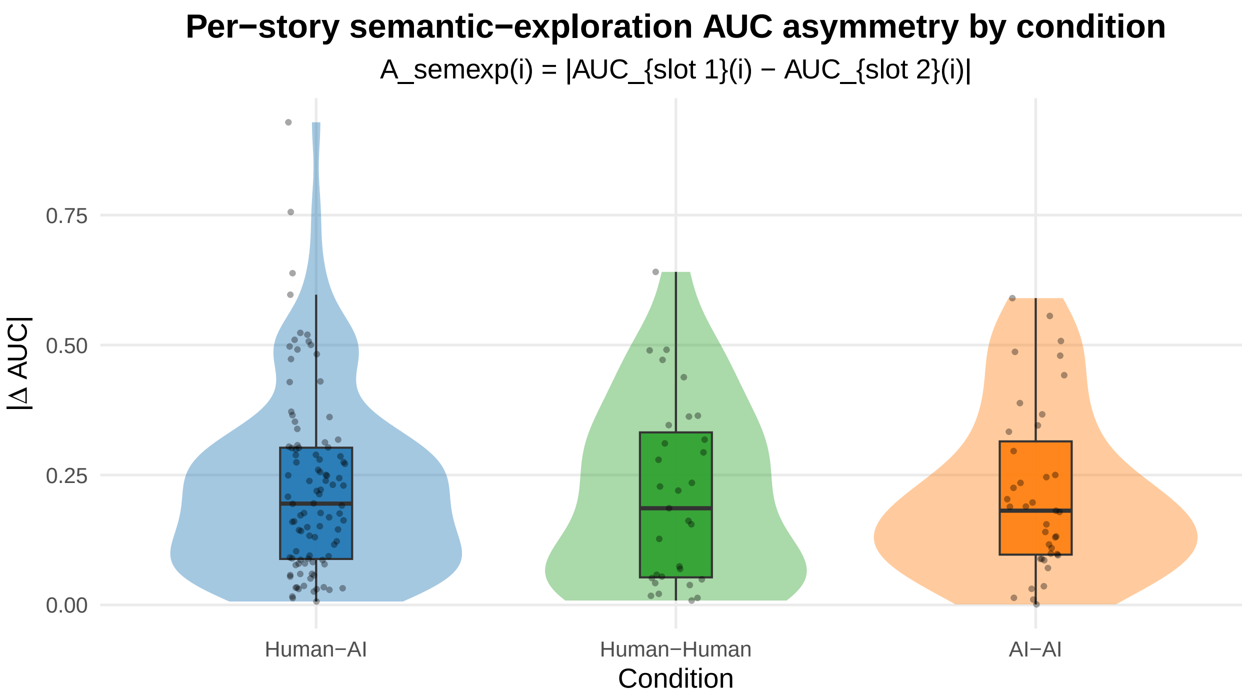


Table 4: Secondary: per-story |log(distance) ~ log(k) slope diff| by condition

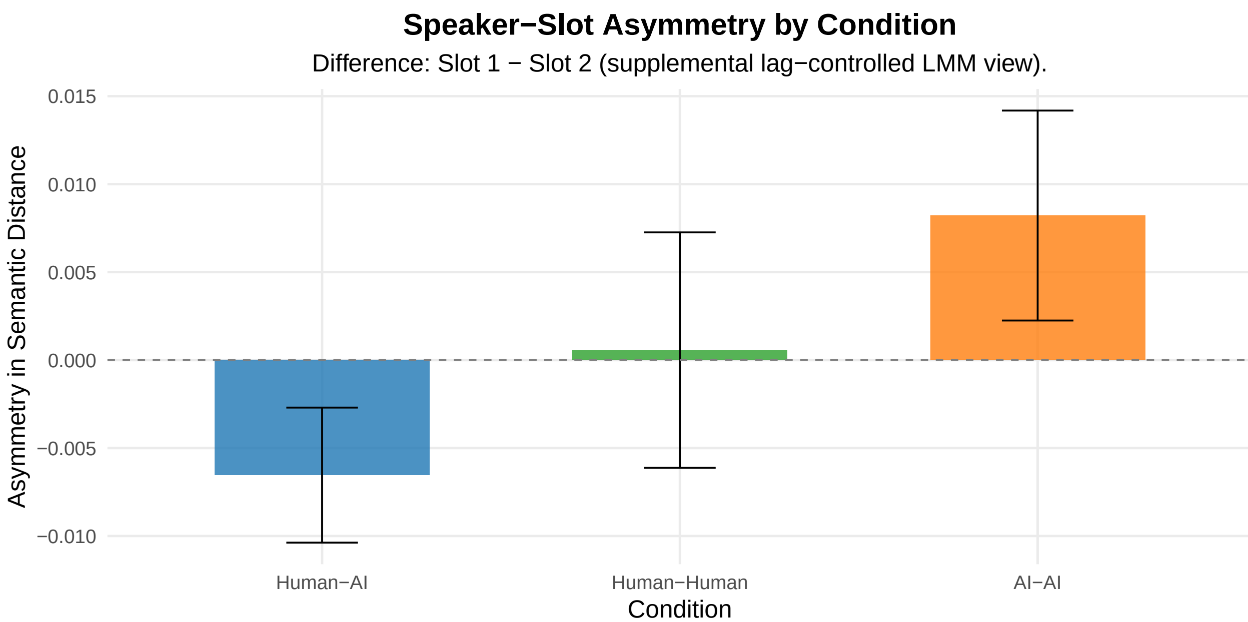
condition	n	mean_A	median_A
Human-AI	100	0.093	0.081
Human-Human	31	0.098	0.083
AI-AI	39	0.078	0.067

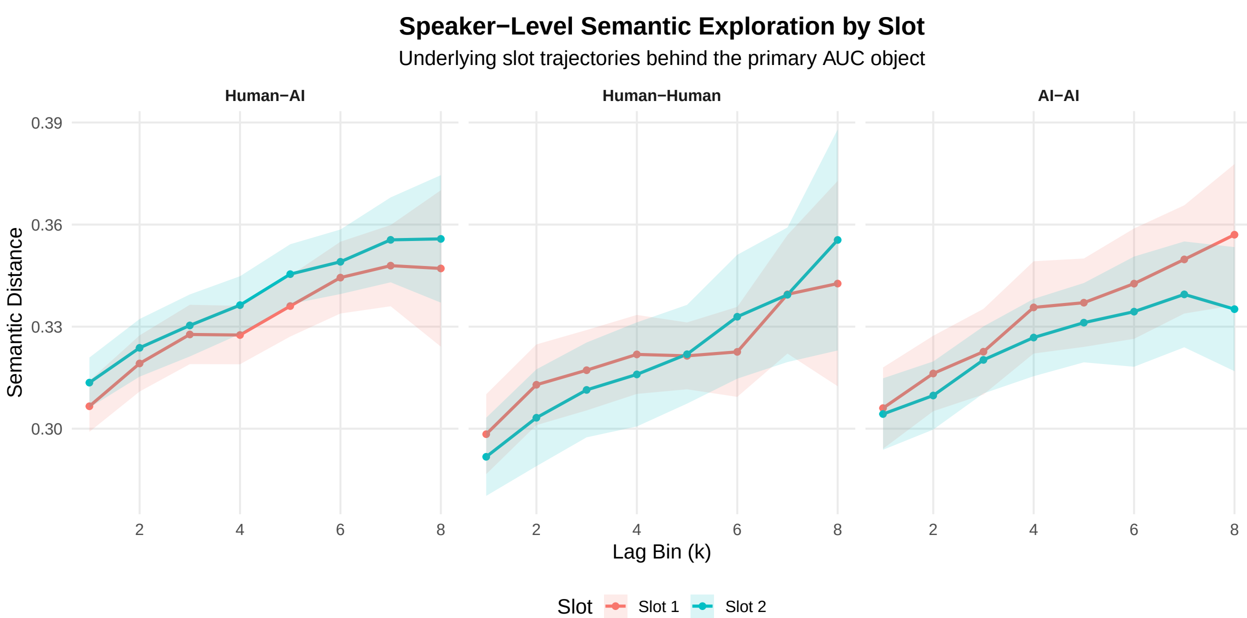
5 Supplemental: Lag-Binned Slot Contrast

```
## Speaker-slot asymmetry model:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: distance ~ k + slot * condition + (1 | conversation_id)
##    Data: df_speaker
##
## REML criterion at convergence: -9257.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.2303 -0.5871 -0.0120  0.5765  5.7695
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0009273 0.03045
##  Residual                    0.0014456 0.03802
## Number of obs: 2628, groups:  conversation_id, 170
##
## Fixed effects:
##                                  Estimate Std. Error         df t value
## (Intercept)                     3.041e-01  3.636e-03  2.786e+02  83.641
## k                               6.318e-03  3.334e-04  2.458e+03  18.951
## slotSlot 2                      6.537e-03  1.958e-03  2.454e+03   3.338
## conditionHuman-Human           -1.051e-02  6.852e-03  1.977e+02  -1.534
## conditionAI-AI                  8.066e-04  6.294e-03  1.979e+02   0.128
## slotSlot 2:conditionHuman-Human -7.103e-03  3.936e-03  2.454e+03  -1.805
## slotSlot 2:conditionAI-AI      -1.475e-02  3.620e-03  2.454e+03  -4.076
##                                Pr(>|t|)
## (Intercept)                     < 2e-16 ***
## k                               < 2e-16 ***
## slotSlot 2                     0.000855 ***
## conditionHuman-Human           0.126682
## conditionAI-AI                 0.898148
## slotSlot 2:conditionHuman-Human 0.071240 .
## slotSlot 2:conditionAI-AI      4.73e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) k      sltSl2 cndH-H cAI-AI sS2:H-
## k           -0.392
## slotSlot 2  -0.269  0.000
## cndtnHmn-Hm -0.445 -0.011  0.143
## condtnAI-AI -0.484 -0.012  0.156  0.260
## sltS2:cnH-H  0.134  0.000 -0.497 -0.287 -0.077
## sltS2:AI-AI  0.146  0.000 -0.541 -0.077 -0.288  0.269
```

Table 5: Speaker-slot asymmetry in semantic exploration by condition

condition	difference	se	t	p
Human-AI	-0.007	0.002	-3.338	0.001
Human-Human	0.001	0.003	0.166	0.868
AI-AI	0.008	0.003	2.699	0.007

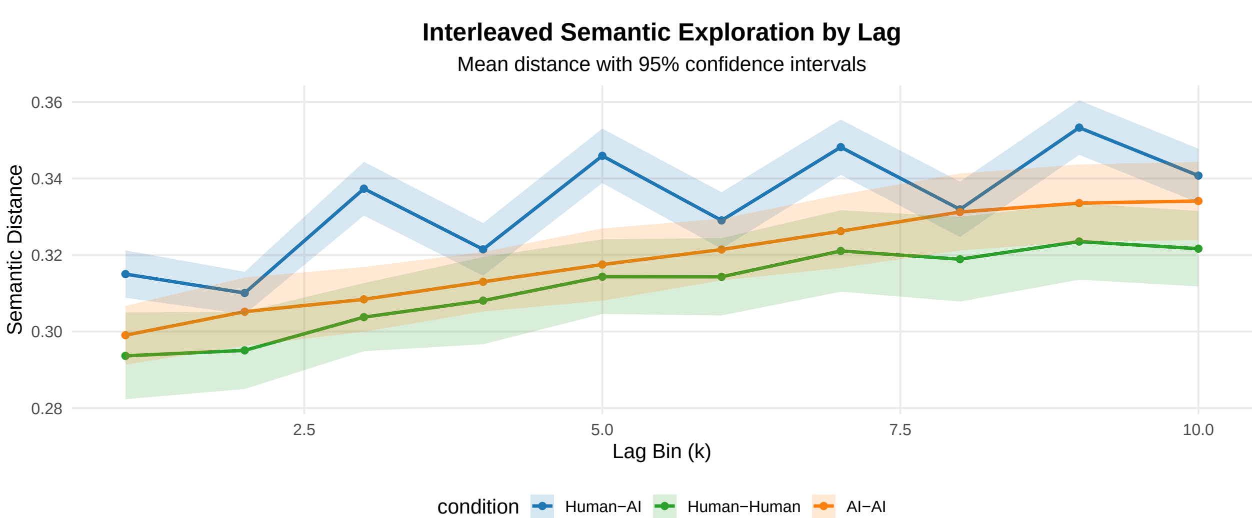




6 Supplemental: General Condition-Level Comparison

Table 6: Interleaved semantic distance by lag and condition

condition	k	mean_distance	sd_distance	n
Human-AI	1	0.315	0.032	100
Human-AI	2	0.310	0.028	100
Human-AI	3	0.337	0.036	100
Human-AI	4	0.321	0.035	100
Human-AI	5	0.346	0.036	100
Human-AI	6	0.329	0.038	100
Human-AI	7	0.348	0.037	100
Human-AI	8	0.332	0.037	100
Human-AI	9	0.353	0.036	100
Human-AI	10	0.341	0.036	100
Human-Human	1	0.294	0.032	31
Human-Human	2	0.295	0.029	31
Human-Human	3	0.304	0.025	31
Human-Human	4	0.308	0.032	31
Human-Human	5	0.314	0.028	31
Human-Human	6	0.314	0.029	31
Human-Human	7	0.321	0.030	31
Human-Human	8	0.319	0.031	31
Human-Human	9	0.324	0.028	31
Human-Human	10	0.322	0.028	31
AI-AI	1	0.299	0.024	39
AI-AI	2	0.305	0.028	39
AI-AI	3	0.308	0.027	39
AI-AI	4	0.313	0.025	39
AI-AI	5	0.318	0.030	39
AI-AI	6	0.321	0.026	39
AI-AI	7	0.326	0.030	39
AI-AI	8	0.331	0.032	39
AI-AI	9	0.334	0.032	39
AI-AI	10	0.334	0.033	39




```
## Interleaved model summary:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

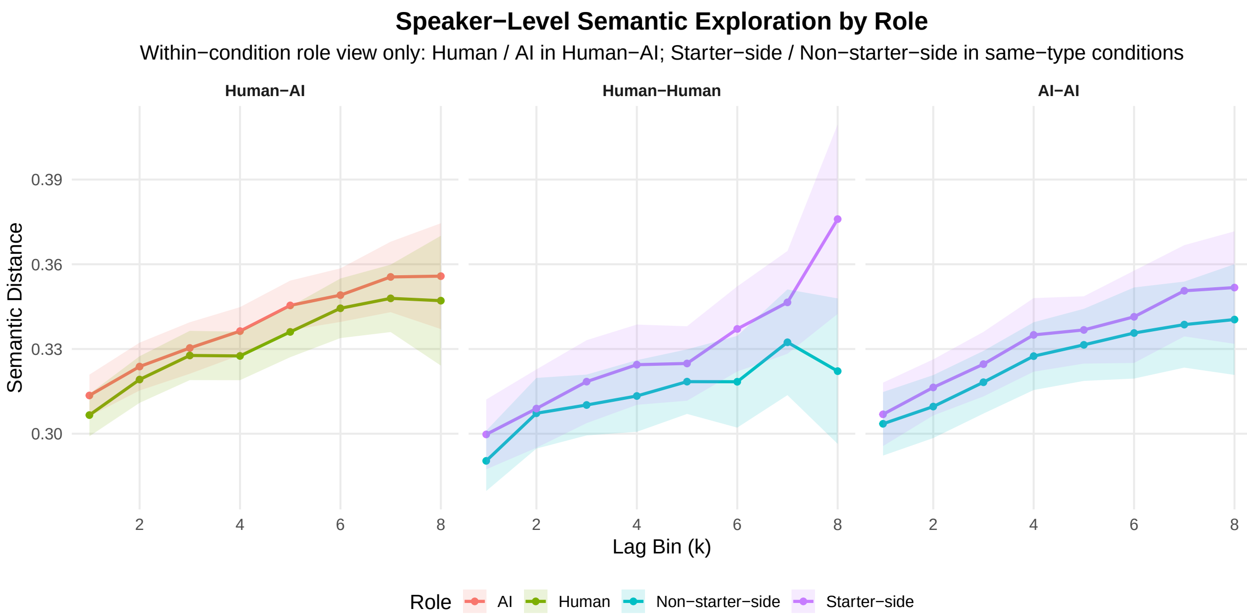
```
## lmerModLmerTest]
## Formula: distance ~ k * condition + (1 | conversation_id)
##    Data: df_interleaved
##
## REML criterion at convergence: -8368.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5669 -0.5987  0.0277  0.6172  3.0302
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0008399 0.02898
##  Residual                    0.0002923 0.01710
## Number of obs: 1700, groups:  conversation_id, 170
##
## Fixed effects:
##                          Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)             3.143e-01  3.125e-03  2.104e+02 100.587  < 2e-16 ***
## k                       3.456e-03  1.882e-04  1.527e+03  18.362  < 2e-16 ***
## conditionHuman-Human   -2.172e-02  6.423e-03  2.104e+02  -3.381  0.00086 ***
## conditionAI-AI         -1.772e-02  5.899e-03  2.104e+02  -3.004  0.00299 **
## k:conditionHuman-Human -2.707e-05  3.869e-04  1.527e+03  -0.070  0.94423
## k:conditionAI-AI        6.156e-04  3.553e-04  1.527e+03   1.733  0.08337 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) k      cndH-H cAI-AI k:cH-H
## k           -0.331
## cndtnHmn-Hm -0.486  0.161
## condtnAI-AI -0.530  0.175  0.258
## k:cndtnHm-H  0.161 -0.486 -0.331 -0.085
## k:cndtAI-AI  0.175 -0.530 -0.085 -0.331  0.258

##
## Condition contrasts (interleaved view):

##  contrast                   estimate      SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)  0.02187 0.00606 167   3.608  0.0012
##  (Human-AI) - (AI-AI)        0.01433 0.00557 167   2.575  0.0327
##  (Human-Human) - (AI-AI)    -0.00754 0.00709 167  -1.063  0.8686
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

7 Supplemental: Role-Level View

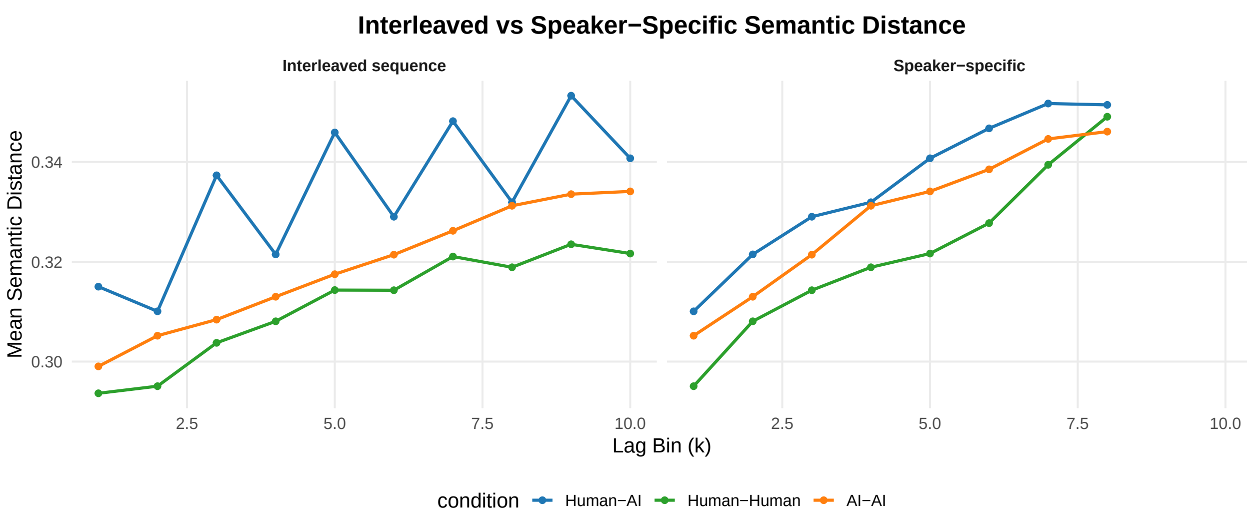
Table 7: Speaker-level semantic distance by condition, role, and lag

condition	condition_role	k	mean_distance	n
Human-AI	AI	1	0.314	100

condition	condition_role	k	mean_distance	n
Human-AI	AI	2	0.324	100
Human-AI	AI	3	0.330	100
Human-AI	AI	4	0.336	100
Human-AI	AI	5	0.345	100
Human-AI	AI	6	0.349	100
Human-AI	AI	7	0.356	99
Human-AI	AI	8	0.356	55
Human-AI	Human	1	0.307	100
Human-AI	Human	2	0.319	100
Human-AI	Human	3	0.328	100
Human-AI	Human	4	0.328	100
Human-AI	Human	5	0.336	100
Human-AI	Human	6	0.344	100
Human-AI	Human	7	0.348	99
Human-AI	Human	8	0.347	55
Human-Human	Non-starter-side	1	0.290	31
Human-Human	Non-starter-side	2	0.307	31
Human-Human	Non-starter-side	3	0.310	31
Human-Human	Non-starter-side	4	0.313	31
Human-Human	Non-starter-side	5	0.318	31
Human-Human	Non-starter-side	6	0.318	31
Human-Human	Non-starter-side	7	0.332	31
Human-Human	Non-starter-side	8	0.322	31
Human-Human	Starter-side	1	0.300	31
Human-Human	Starter-side	2	0.309	31
Human-Human	Starter-side	3	0.318	31
Human-Human	Starter-side	4	0.324	31
Human-Human	Starter-side	5	0.325	31
Human-Human	Starter-side	6	0.337	31
Human-Human	Starter-side	7	0.347	31
Human-Human	Starter-side	8	0.376	31
AI-AI	Non-starter-side	1	0.304	39
AI-AI	Non-starter-side	2	0.310	39
AI-AI	Non-starter-side	3	0.318	39
AI-AI	Non-starter-side	4	0.327	39
AI-AI	Non-starter-side	5	0.331	39
AI-AI	Non-starter-side	6	0.336	39
AI-AI	Non-starter-side	7	0.339	39
AI-AI	Non-starter-side	8	0.340	39
AI-AI	Starter-side	1	0.307	39
AI-AI	Starter-side	2	0.316	39
AI-AI	Starter-side	3	0.325	39
AI-AI	Starter-side	4	0.335	39
AI-AI	Starter-side	5	0.337	39
AI-AI	Starter-side	6	0.341	39
AI-AI	Starter-side	7	0.351	39
AI-AI	Starter-side	8	0.352	39



8 Supplemental: Interleaved vs Speaker View



9 Summary

Primary cross-condition result: per-story $A_{\text{semexp}} = |\text{AUC}_{\text{slot 1}} - \text{AUC}_{\text{slot 2}}|$ from trapezoidal integration of the distance(k) curve over the intersected k range. Tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon. A secondary log-distance-slope asymmetry is reported for dynamic-shape continuity with the paper.

Supplemental views retained:

- lag-controlled slot-contrast LMM (`distance ~ k + slot * condition`)
- interleaved sequence model and condition descriptives
- within-condition role curves (Human / AI in Human-AI; Starter-side / Non-starter-side in same-type)
- interleaved vs speaker-specific side-by-side view

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized AUC asymmetry across all conditions)
- Human-AI decomposition (`speaker_type \u00d7 speaker_is_starter` within Human-AI only)

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